

Baseline, Linear, and Flexible Models on a Synthetic Regression Task

A short tutorial report generated from `metrics.json`

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Abstract

We compare a constant mean baseline, an ordinary least-squares line, and a degree-9 polynomial on a one-dimensional synthetic regression dataset. The polynomial attains the lowest training error but the highest held-out error, a compact illustration of overfitting. All figures are single-split results and not uncertainty estimates.

Method

Data: $y = 2x + 1 + \epsilon$ with x evenly spaced on $[-1, 1]$, Gaussian noise $\sigma = 0.3$, seed 42, 40 points. We fix a 20/20 train/held-out split with a seed-42 permutation. Each model is fitted on the 20 training points only and scored by mean squared error (MSE) on the 20 held-out points.

Results

Model	Training MSE	Held-out MSE
Mean baseline	1.2922	1.7700
Linear	0.0289	0.1035
Degree-9 polynomial	0.0223	0.1429

Discussion

The polynomial fits the training rows best (0.0223 versus 0.0289) yet generalizes worst (0.1429 versus 0.1035): extra flexibility bought a small training gain at the cost of held-out accuracy. Because the evaluation uses a single split, these numbers do not quantify uncertainty, and the held-out set should not be used for model selection. Values were copied from `metrics.json` at four decimal places and contain no new measurements.